

How a Brain-Body Network Detects Its Own Emotional State Changes

Tonic Context, Phase-Wave Differentials, and Distributed Self-Registration in Self-Aware Networks

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

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Abstract

An emotional episode is accompanied by changes distributed across the brain and body, yet no single measurement is the emotion and no separate inner observer is available to read those measurements. This paper asks a mechanistic question: how can a distributed brain-body system detect a transition in its own condition and use that transition in feeling, regulation, choice, and action? The proposal is developed from the 2011-2026 genealogy of Self-Aware Networks (SAN). Early Neo Mind Cycle records described a senses-brainwaves-action loop and an external neurofeedback mirror in which selected electroencephalographic changes were transformed into light and sound, returned through the senses, related to thoughts and feelings, and used for learned self-discrimination. Owner-dated 2012 notes then made emotion prediction-based, connected dynamic electrical expectation to brain-body-sense feedback, distinguished the physical form of a stimulus from its learned meaning, and described distributed observation without a singular observer. A 2017 public formulation supplied both a multimodal measurement program and a crucial boundary: thoughts and feelings are reflected in brainwaves, not identical to them. A 2018 public article described feelings and emotions as patterns with temporal profiles and spatial shape. Public 2018-2020 Neural Lace episodes then extended the program from multimodal emotion inference to the limits of simple decoding and to synchronized, longitudinal, person-level measurement. Service-dated 2021 recordings made emotional change explicitly baseline-relative across neuron, microcircuit, whole-brain, and body scales and joined tonic / phasic self-discrimination, causal self-detection, and affective geometry. A public 2022 Git sequence then joined detectable departures from regular oscillatory relations to receiver change, tonic and phasic context, bodily location, rendered feeling, valuation, action, and sensory return; two public 2022 videos independently fixed the partial-projection, distributed-observer, tonic-context, and rendered-emotion formulation. Later SAN work generalized the external loop into an intrinsic account in which interoceptive receptors, neural receivers, recurrent circuits, and bodily feedback are altered by the state they register. The maintained condition supplies context; receiver-relative departures supply new differences; their propagation changes the system's subsequent processing. Only after this operation is defined is the later term Phase-Wave Differential (PWD) introduced for a multivariate difference in timing, phase, magnitude, duration, waveform, or spatial distribution relative to a receiver and its expected state. A formal partially observed dynamical model separates latent brain-body state, measured projections, reported feeling, psychological valence, effective electrical drive, and action. The resulting experimental program compares person-specific and cross-person models, unimodal and multimodal estimates, and veridical, delayed, replayed, decorrelated, phase-scrambled, and sham feedback. SAN gains support only if its receiver-relative variables add held-out prediction and selective causal leverage beyond conventional physiological, spectral, behavioral, movement, and context baselines.

Self-Aware Networks, brain-body dynamics, Phase-Wave Differential, tonic context, neural rendering, affective neuroscience

Keywords: emotion, interoception, psychophysiology, neurofeedback,

Scope and evidence boundary

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1. The familiar event

Imagine that you are angry and have five minutes before an important interview. The anger is not merely the sentence *I am angry*. Your breathing may change. Your heart may beat differently. Your facial and postural muscles may remain prepared for confrontation. Attention may return to the provoking event. A possible future may be rehearsed repeatedly. The same question may feel more urgent, the same face more threatening, and the same action more available than it did ten minutes earlier.

You may also notice this transition. Sometimes the change is detected through a heartbeat, heat, breath, tension, trembling, nausea, pressure, or an impulse to move. Sometimes the first accessible sign is cognitive: a narrowing of attention, a repeated memory, or the changed meaning of the room. Sometimes another person notices the change before you do.

The ordinary event therefore contains several different problems:

1. What changed physically?
2. Which parts of that change reached measurement or awareness?
3. How did the system distinguish the new condition from its preceding condition?
4. How did the change alter what could be remembered, selected, said, or done?
5. How did the consequence of the action modify the next state?

Calling the event an *emotion* names the phenomenon. It does not yet explain how the system detects its own transition.

2. The scientific problem

Emotion research has no shortage of measurements. Electroencephalography (EEG) measures scalp potential differences produced by many underlying current sources. Electrocardiography and photoplethysmography measure aspects of cardiac dynamics. Electrodermal activity measures changes related to sweat-gland activity. Electromyography measures muscle activity. Respiration, temperature, pupil behavior, gaze, movement, hormone concentrations, hemodynamics, and self-report reveal other projections of the changing organism.

These measurements are neither interchangeable nor complete. A change in arousal is not automatically a change in pleasantness. A facial-muscle signal is not a whole feeling. A classifier label is not the event itself. A high confidence estimate may still reflect movement, task structure, respiration, identity leakage, or repeated trials from the same person.

Direct stimulation supplies an especially useful warning. Human amygdala stimulation can produce reliable autonomic changes without reliably producing a corresponding reported emotion [1]. Physiology can therefore change without the full subjective event. Conversely, similar reports can arise from different detailed trajectories. The explanandum is not a one-signal fingerprint. It is a distributed transition and the organism's registration and use of that transition.

The second problem is architectural. If the brain contains no miniature person watching an inner display, then the measurement cannot be delivered to a hidden reader. The participating system must perform the detection. A receptor changes when its ligand, force, voltage, temperature, or chemical environment changes.

A receiving neuron changes when current arrives under its present membrane, dendritic, inhibitory, and synaptic conditions. A recurrent circuit changes when its inputs, coupling, and bodily return change. At every scale, detection is a causal alteration of the detector.

The central question can now be stated without SAN terminology:

How can many partial receivers jointly register a changing brain-body condition, distinguish it from the condition they were maintaining, and use the difference to modify feeling, prediction, regulation, and action?

3. Established evidence: emotion is a distributed brain-body transition

3.1 Dynamic physiology, not one static signature

Classic experiments found distinguishable patterns across heart rate, skin conductance, temperature, respiration, and somatic activity during emotional conditions [2]. Later work showed that short-lived multichannel autonomic states emerge, dissipate, and recur during emotional episodes and can also appear at lower magnitude during rest [3]. The important unit is therefore not always a single average or one channel. It can be a trajectory through a multivariate state space.

Brain-heart analyses similarly report reciprocal and time-dependent interaction. In one computational analysis of EEG and cardiac data, ascending autonomic modulation preceded aspects of neural dynamics during emotional arousal, after which brain and body continued to influence one another [4]. Whatever the final causal interpretation of a particular analysis, a one-way brain-to-body diagram is insufficient.

3.2 The organism senses parts of its own condition

Interoception names the nervous system's sensing and integration of internal bodily conditions. Heartbeat-detection performance has been associated with insula activity, reported visceral awareness, and aspects of emotional experience [5]. This does not imply that heartbeat counting is a complete measure of interoception or emotion. It demonstrates that measurable individual differences in access to bodily signals relate to neural and subjective variables.

A 2025 physiology-based study operationalized affective awareness as the within-person relationship between facial electromyography and reported affect. That relationship was positive on average, differed reliably between people, and was stable over time [6]. This is important for the present proposal: self-detection is expected to be person-relative before it is expected to generalize across people.

Bodily location is also measurable as a report dimension. In five experiments, participants marked body regions whose felt activity increased or decreased during emotion-eliciting words, stories, films, or faces; the resulting maps showed statistically separable structure across emotion conditions [84]. Those maps are reports of bodily sensation, not direct neural source maps and not a complete identity for emotion. They nevertheless establish a conventional baseline against which the later SAN affective-geometry proposal can be tested.

3.3 Feedback changes the system being measured

In neurofeedback, a measured variable is transformed into a stimulus and returned to the participant. The feedback is not a neutral report. It becomes a new sensory event, changes expectation and engagement, and may change the measured variable on the next cycle. Real-time fMRI and simultaneous EEG-fMRI studies

show that participants can alter selected measures during emotion-regulation tasks under bounded protocols [7-9].

The causal interpretation depends on controls. In a 2026 virtual-reality study, sham feedback altered alpha power during an attentional task even though the feedback was not contingent on the participant's actual brain activity [10]. Expectancy, task engagement, feedback valence, and sensory stimulation can therefore change the target measure. A useful experiment must compare true contingency with delayed, replayed, decorrelated, and sham conditions rather than assuming that a changing signal proves self-regulation of the named neural source.

3.4 Physical form, learned meaning, and receiver history are separable

An emotional stimulus is not exhausted by either its physical form or its learned meaning. A voice has duration, intensity, pitch, rhythm, and timbre before a listener identifies its words. A familiar word can acquire a learned meaning that an acoustically similar pseudoword lacks. An angry tone can alter attention and autonomic state even when its sentence is semantically neutral. Conversely, emotionally meaningful words can change appraisal even when their prosody is controlled.

These components can be separated experimentally. Zora and colleagues compared words with acoustically matched pseudowords and crossed linguistic with affective prosody. They found distinguishable earlier responses to linguistic and affective changes and a later response when emotional semantics and affective prosody were integrated [91]. This was a small Swedish ERP study: sixteen people were enrolled and one was excluded from analysis. Carbone and colleagues studied seventy-seven native French listeners while crossing emotional semantics and prosody and measuring judgments and skin conductance. Prosody had the larger behavioral effect, while the autonomic effect of angry prosody was limited to the female participants, illustrating both component separation and person-dependence [92]. These studies establish useful experimental dissociations; their sample and subgroup bounds do not establish a universal neural code.

This distinction matters for self-registration. The same acoustic waveform can have different consequences for two listeners with different language, learning, and bodily histories. The same learned meaning can arrive through different physical forms. A mechanism therefore needs at least three layers:

physical stimulus structure

-> receiver-specific transformation shaped by learning and context

-> distributed neural, autonomic, felt, and behavioral consequence

The consequence can begin before conscious identification, but it is not therefore independent of the receiver. The receiver's learned structure is part of the causal operation.

4. The operation in plain language

Before introducing the SAN terms, the proposed operation can be stated in nine steps.

1. A perception, memory, thought, prediction, drug, bodily change, or social event alters parts of the organism.
2. Neural currents, firing, coupling, neuromodulation, cardiac dynamics, breathing, muscle state, temperature, movement, and other variables change on different time scales.
3. No receiver obtains the whole transition. Each receptor, cell, circuit, and instrument receives a partial projection.

4. What the projection does depends on the receiver’s current state, recent history, anatomy, timing, and expectation.
5. Internal bodily signals and recurrent neural signals return portions of the transition to the system.
6. The returning signals change the receiving system. That causal change is the physical act of registration.
7. Many local registrations alter a distributed body-self-world model and the availability of memories, interpretations, and candidate actions.
8. One or more actions change the body and environment.
9. The returned consequences change the next cycle and, through learning, later cycles.

There is no point at which a hidden observer must be inserted. The system detects the transition because the transition changes the system.

The transition also has a shape in the ordinary, measurable sense. Anger may build quickly and decay slowly. Anxiety may be reported in the chest and stomach, while another episode is experienced as pressure in the head or as whole-body agitation. A feeling can change location, spread, contract, recur, or become linked to a particular posture, memory, object, or route of action. This paper calls that bundle its **affective geometry** only after the component measurements are named: onset, duration, rise and fall, magnitude, body-location map, neural pathway distribution, and cross-modal coupling. The phrase does not declare a new substance. It summarizes a time-varying measurement target.

5. Formal model

5.1 A partially observed brain-body state

Let $x_p(t)$ be the latent, person-specific brain-body state of participant p . It includes variables at multiple scales and is not assumed to be recoverable in full. Let $u^{\text{stim}}(t)$ be the measurable physical structure of an event or sensory input, including any external feedback after sensory re-entry. Let $\mathcal{H}_p^{\text{learn}}(t)$ represent the participant’s relevant learned history and $c_p(t)$ the current context. The effective event for that receiver is not assumed to equal the raw signal. It is a person- and context-dependent transformation $e_p(t)$. Let $a(t)$ be action, $b(t)$ bodily and environmental return, and $\eta(t)$ unresolved dynamics:

$$\begin{aligned} e_p(t) &= E_p\left(u^{\text{stim}}(t), \mathcal{H}_p^{\text{learn}}(t), c_p(t)\right), \\ x_p(t + \Delta t) &= F_p(x_p(t), e_p(t), a(t), b(t), \eta(t)). \end{aligned} \quad [1]$$

Equation [1] leaves open how much of a response is explained by acoustic, visual, or other physical structure; how much by learned meaning; and how much by their interaction. Those components must be crossed experimentally rather than assigned from a label after the fact.

Sensor i returns only a transformed projection:

$$z_{i,p}(t) = H_i(x_p(t), c_p(t), m_i(t)) + \epsilon_i(t), \quad (2)$$

where $c_p(t)$ is context, $m_i(t)$ includes movement and measurement nuisance, and $\epsilon_i(t)$ is unresolved error. Equation [2] forbids a classifier from silently treating EEG, heart rate, facial muscle, or self-report as the whole emotional state.

5.2 A receiver-relative difference

Receiver j has a state $r_j(t)$, a recent history $h_j(t)$, and an input relation $q_j(t)$. Because phase is circular, pathways may be graph-valued, and the components have different units, the expected relation is not represented by one Euclidean mean. It is a conditional probability distribution over a mixed continuous, circular, waveform, and graph-valued state. For every measurable set B :

$$\mathcal{P}_{j,p,t}(B) = \Pr(q_j(t) \in B \mid r_j(t), h_j(t), c_p(t)). \quad (3)$$

Where a density or probability mass exists it may be estimated, but Equation [3] does not require the continuous components of q_j to take one exact value with nonzero probability.

The arriving relation is represented as a vector rather than a single phase:

$$q_j(t) = [A_j(t), \phi_j(t), f_j(t), d_j(t), w_j(t), \ell_j(t)], \quad (4)$$

where A is a declared magnitude, ϕ phase, f frequency, d duration, w waveform features, and ℓ spatial or pathway location. A component-wise discrepancy is then defined:

$$\delta_j(t) = [z_A(A_j), d_{\mathbb{S}^1}(\phi_j, \mathcal{P}_j), z_f(f_j), z_d(d_j), D_w(w_j, \mathcal{P}_j), D_G(\ell_j, \mathcal{P}_j)]_t. \quad (5)$$

Here the z terms are residuals standardized using training data only; $d_{\mathbb{S}^1}$ is a preregistered circular phase discrepancy; D_w is a declared waveform distance; and D_G is either graph distance or a preregistered route-embedding distance. No subtraction of categorical pathway labels is permitted.

A single receiver is insufficient to represent stable alignment, interference, or structured dispersion. For receiver pair j, k , define relational features and a population circular-variance term:

$$\begin{aligned} C_{jk}(t) &= [\cos(\phi_j - \phi_k), \sin(\phi_j - \phi_k), \hat{\rho}_{jk}^A(W_t), \widehat{\Delta\tau}_{jk}(W_t)], \\ V_{\mathcal{R}}(t) &= 1 - \left| \frac{1}{|\mathcal{R}|} \sum_{j \in \mathcal{R}} e^{i\phi_j(t)} \right|. \end{aligned} \quad (6)$$

W_t is a preregistered analysis window centered or ending at t . $\hat{\rho}_{jk}^A(W_t)$ is the declared within-window amplitude co-variation estimator, and $\widehat{\Delta\tau}_{jk}(W_t)$ is the within-window relative-delay estimator. C_{jk} therefore preserves pairwise phase relation, amplitude co-variation, and relative delay without pretending that correlation or lag is defined at one instantaneous sample. $V_{\mathcal{R}}$ distinguishes tighter phase concentration from structured dispersion or splay. Neither alignment nor dispersion is declared meaningful in isolation; its consequence depends on receiver, pathway, task, and time.

The recovered 2018 and 2022 sources motivate an additional declared view of the same variables rather than a new latent essence. For episode e , its affective geometry is the preregistered bundle of transition onset, peak, duration, rise and decay, body-location distribution, pathway distribution, and cross-modal coupling. Each component must be measured separately; a verbal body map is not silently substituted for a neural source map. A coefficient-of-variation statistic may be tested as one normalized measure of dispersion relative to a collective mean [83], but it has no privileged status: it must outperform circular variance, entropy, synchrony, power, rate, and history-aware baselines on held-out data. This makes the older spatial-temporal claim falsifiable without pretending that every feeling has one fixed shape.

5.3 Distributed self-registration

Let T_j be the receiver's state transition caused by δ_j :

$$r_j(t + \Delta t) = T_j\left(r_j(t), \delta_j(t), u_j^{\text{local}}(t)\right), \quad (7)$$

Here $u_j^{\text{local}}(t)$ is distinct from the external or sensory structure $u^{\text{stim}}(t)$ in Equation [1]. It includes local inhibition, neuromodulation, metabolism, and other inputs acting at receiver j . The system has physically registered a difference when that difference changes a receiver and the change has a downstream consequence. Physical registration does not by itself imply felt emotion or report.

The 2021 genealogy explicitly spans neurons, microcircuits, larger circuits, whole-brain dynamics, and the body. Registration is therefore retained first as a multiscale object:

$$\mathcal{S}_{p,s}(t) = \left\{ \delta_j(t), C_{jk}(t), \Delta r_j(t), o_j^{\text{pre}}(t) \right\}_{j,k \in \mathcal{R}_s}, \quad (8)$$

$s \in \{\text{cell, circuit, system, body}\}.$

Δr_j is the measured receiver-state change and o_j^{pre} is a preregistered downstream observable chosen without using the test outcome. This prevents outcome leakage from being smuggled into the definition of registration. If a scalar summary is needed, it must preserve declared signed and relational components:

$$S_p(t) = \sum_s \alpha_{s,p} w_{s,p}^T \text{whiten}_{\text{train}}(\text{vec}_{\text{fixed}} \mathcal{S}_{p,s}(t)). \quad (9)$$

The vectorization uses a fixed, preregistered feature ordering. All scaling, covariance, weights, and thresholds are fitted inside training folds. S_p is not consciousness or emotion by definition. It is an optional out-of-sample summary of a declared multiscale registration object.

5.4 Feeling, report, and action remain distinct

Physical registration, broader integration, accessible felt change, explicit awareness, verbal report, and action are not one endpoint. Define the participant-level multiscale predictor $\mathcal{S}_p(t) := \text{concat}_s \text{vec}(\mathcal{S}_{p,s}(t))$, preserving scale labels rather than silently pooling them. For each declared target m , fit a separate observation model:

$$P\left(y_p^{(m)}(t) \mid z_{1:n,p}(t), \mathcal{S}_p(t), c_p(t), h_p(t)\right), \quad [10]$$

$m \in \{\text{integration, felt arousal, felt valence, category, awareness, report, readiness, regulation}\}.$

Action candidate k has accessibility $g_{k,p}(t)$, learned value $v_{k,p}(t)$, and effective drive $D_{k,p}(t)$:

$$K_p(t) \sim \pi_p(k \mid x_p(t), g_{k,p}(t), v_{k,p}(t), D_{k,p}(t)), \quad (11)$$

$a_p(t) = A_p(K_p(t)).$

Psychological valence contributes to v_k . Coherence-weighted electrical drive can contribute to D_k . They interact but are not the same variable. The detailed definition of D_k belongs to the companion motor-choice paper; here it means a separately measured, weighted contribution of phase-compatible synaptic and

population inputs to a candidate circuit. Muscular energy is supplied later by neuromuscular transmission, calcium-dependent cross-bridge cycling, and ATP; a neural timing pattern recruits that machinery rather than replacing its energy source.

The consequence updates the model:

$$(x_p, v_p, g_p, \hat{\mathbf{q}}_p)_{t+\Delta t} = U_p(x_p, v_p, g_p, \hat{\mathbf{q}}_p, a_p, o_p - \hat{o}_p)_t. \quad (12)$$

Here $\hat{o}_p$ is predicted consequence and o_p observed consequence. $\hat{\mathbf{q}}_p$ is the declared participant-level forecast assembled from the receiver-specific predictive distributions in Equation [3]; its aggregation rule is fitted within training data and cannot use the held-out outcome. Fast receiver-state changes, slower value and accessibility updates, and still slower structural plasticity must be estimated on separate declared time scales.

5.5 The external feedback loop

An external neurofeedback system estimates a target, chooses a bounded display, and returns that display after a measurable latency:

$$\begin{aligned} \hat{y}_p(t) &= G(z_{1:n,p}(t), c_p(t), \theta_p), \\ f_p^{\text{ext}}(t) &= \Pi(\hat{y}_p(t), \sigma_p(t), L, \lambda_p(t)), \\ e_p^{\text{sens}}(t + \lambda_p(t)) &= R(f_p^{\text{ext}}(t)). \end{aligned} \quad (13)$$

Here θ_p is the calibrated model, σ_p its uncertainty, and L safety limits. The mapping R and event-specific end-to-end latency $\lambda_p(t)$ are measured and recorded, not assumed. Because e_p^{sens} becomes a new input in Equation [1], the measurement loop changes the process it observes.

5.6 A SAN-specific success criterion

The proposal earns distinct support only if receiver-relative variables improve held-out performance beyond strong conventional baselines by a preregistered minimum amount:

$$\begin{aligned} \Delta_{\text{SAN}} &= \mathcal{U}(M_{B+S}) - \mathcal{U}(M_B), \\ \text{LCB}_{1-\alpha}(\Delta_{\text{SAN}}) &> \delta_{\min}. \end{aligned} \quad (14)$$

Here B includes anatomy, firing rate, ordinary spectral features, autonomic variables, respiration, movement, task, context, history, and flexible recurrent models; S contains preregistered SAN receiver-relative features; and $\mathcal{U}$ is direction-consistent, so larger always means better. All model and feature selection occurs inside nested training folds. The lower confidence bound must exceed $\delta_{\min}$ in a preregistered analysis and then survive an independent replication.

Prediction alone is insufficient. Let $Y^{(0)}$ be the potential outcome under the matched control, $Y^{(I)}$ under a selective disruption, and $Y^{(I,R)}$ under disruption plus matched rescue:

$$\begin{aligned} \text{LCB}_{1-\alpha}(s_I E[Y^{(I)} - Y^{(0)}]) &> \delta_I, \\ \text{CI}_{1-\alpha}(E[Y^{(I,R)} - Y^{(0)}]) &\subseteq [-\epsilon_R, \epsilon_R], \end{aligned} \quad (15)$$

Equation [15] is the causal gate: disrupt the proposed component in the predicted direction $s_I \in \{-1, +1\}$, fixed before observing outcomes, then restore equivalence to the matched control with the entire confidence interval inside a predeclared rescue margin. The design must justify consistency, positivity, exchangeability or the relevant randomized alternative, interference assumptions, and control of direct sensory or stimulation effects. Nuisance variables cannot be restored selectively in the rescue and then credited to the SAN component.

6. Only now: the SAN terms

The comparatively maintained, recurring condition is called the **tonic context**. It need not be one slow band, one global rhythm, or a perfectly stationary baseline. It is the multiscale condition against which a receiver's incoming relation is interpreted.

A meaningful event-related departure is **phasic** relative to that context. A phasic change need not be universally high-frequency. Its defining property is its event-relative difference and its effect on receivers.

The multivariate receiver-relative difference in Equation [5] is the operation SAN calls a **Phase-Wave Differential (PWD)**. The name emphasizes that the relevant message is not exhausted by a spike count or an absolute phase. It is a change in a wave-related signal relative to the receiver's state, expectation, and route, potentially involving phase, timing, magnitude, duration, waveform, frequency, and spatial distribution.

Equation [6] adds relations among receivers so that phase alignment and structured dispersion are not collapsed into one unsigned amount. Equation [8] keeps cell, circuit, system, and body scales visible before any scalar summary is attempted. These additions matter for emotion because the relevant transition may be a change in coordination, separation, pathway, or bodily return rather than a uniform rise or fall in one band.

When many such local registrations change the distributed body-self-world model, SAN calls the ongoing construction **Neural Rendering**. Rendering does not require a central screen. Each participating population receives partial conditions, transforms them, and changes downstream populations. The changing network is simultaneously the medium, the transformation process, and the distributed observer operation.

In the emotion-specific case:

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tonic brain-body context
-> emotion-relevant transition
-> many receiver-relative differences
-> recurrent and interoceptive return
-> distributed change in the rendered body-self-world state
-> feeling, valuation, regulation, and candidate action
-> returned consequence and learning

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7. The SAN genealogy: from an external mirror to intrinsic self-registration

The present theory did not begin with a new acronym. It developed from an experimental problem.

7.1 2011: the physical loop

SAN's early ancestry described a learning system in which senses changed brainwave activity, brain activity contributed to movement and sound, and those consequences returned through the senses. The recurring relation was called a learning loop that learns itself [11,12].

A second owner-dated July 2011 record already places triggered feelings, thoughts, and actions inside a returning senses-cortex-hypothalamus pathway and describes the experienced self as a collective of cells [60]. This was not yet a formal emotion-specific measurement theory, but it makes the affective ancestry of the physical loop explicit:

sensation -> changing neural activity -> action -> changed sensation

The early notes also treated converging information as adding weight to a candidate action until a threshold was crossed [13]. That was an ancestor of later choice and effective-drive formulations, not yet their mature mechanism. An adaptive EEG / light / sound note from the same period also supplies the external feedback ancestor: the displayed stimulus was meant to change as the measured state changed, so the consequence of one cycle became part of the next cycle [31]. None of these internally dated records is presented as a 2011 public Git publication. Their older dates are owner dates; the immutable public Git fixations begin in 2022.

7.2 2012: the external mirror of felt-state change

Neo Mind Cycle transformed selected EEG changes into light and sound. One 2012 source contrasted a rigid software decoder with a second method in which the measured pattern was returned to the person and decoded by the person's own brain. The loop was described as a high-tech mirror and as expanded self-awareness [14].

Another record reported a fusion of feelings, emotions, internal visualization, and audio feedback representing measured brain activity [15]. The most complete early account described the person gradually noticing that light and sound changed with internal state, thought, feeling, and representation; learning cross-modal associations; changing the state; acting; and observing the returned consequence [16]. A later 2012 note ordered the operation into measurement, transformation, sensory return, changed state, recognition, learned association, and regulation [17].

Two companion notes broaden the operation. A June 2012 source describes the cortex as learning a model of the body and behavior and asks how that predictive model could become a form of self-awareness [32]. A July 2012 source connects individual preferences, reinforcement, recognition of inner state, and later choice [33]. These are partial ancestors. They do not yet supply the later interoceptive, tonic / PWD, or person-relative formalism.

The exhaustive corpus pass recovered four additional 2012 steps. An owner-dated April note states directly that emotion is a feeling based on predictions [64]. An embedded June formulation describes awareness as a dynamic electrical expectation and self-awareness as a feedback loop among brainwaves, body movement, and incoming senses [65]. An August note treats EEG variables as partial projections of present emotional and sensory state and proposes that repeated feedback could teach larger-scale concepts of internal coordination and self [66]. A second August note replaces a singular observer with many distributed predictions and includes feelings among the contents of that prediction-and-memory system [67].

A December 2012 discussion adds a previously omitted receiver-history atom. In explicitly labeled Micah turns, words and stories are described as producing physical neural consequences and sometimes initiating an emotional transition, while unfamiliar language, raw sound, prosody, imagery, conscious recognition, and learned meaning are treated as distinguishable influences [85]. The historical wording is exploratory and includes claims broader than the present paper requires. Its durable contribution is narrower: a stimulus can affect a receiver through physical structure, learned semantics, or their interaction, and the effect depends on the receiving system's history. The source is internally dated December 10, 2012; its first immutable public Git fixation is June 8, 2022.

These sources materially strengthen continuity, but their dates require two separate statements. The wording

is owner- or source-dated to 2012; the immutable public Git fixation is in 2022. They therefore establish an owner-dated conceptual genealogy and a later public custody date, not a fabricated 2012 Git publication.

These records establish a motivating observation, not a clinical trial. The instrument displayed a selected correlate mixed with artifacts and nuisance variables. It did not display an emotion in full. Nevertheless, the operation posed the enduring question: if a person can learn a partial external rendering of a changing internal condition, how does the unassisted nervous system register its own changing condition?

7.3 2013-2015: prediction, distributed detection, and action

The next sources supplied missing parts. Fear was described as a remembered causal relationship projected into a possible future, while accumulating evidence and learned criteria contributed to action [18]. A 2013 no-inner- observer account described cells collecting and transforming information with no additional witness [19]. A 2015 source related conscious feeling to a self-referential network predicting body sensations and movements [20]. Another 2013 note joins a cell-level transition to an action experienced as the system's own [34]. The source a0465z joins distributed cell groups, multisensory self-patterns, body-related feeling, feedback, and action [56]. Its wording is publicly fixed in Git in 2022, but its claimed 2014 source date remains unverified. The ledger therefore marks it source-gated and does not use it as a 2014 priority anchor.

An owner-dated November 2013 source adds a dynamic-self bridge. In Micah-labeled turns, the self is described as a live spatial pattern rebuilt from millisecond coincidence relations; new information changes firing criteria and therefore the system's next action [61]. This is a pre-PWD connection among coincidence, a continuously reconstructed self, and action. Peter Tse's criterial-causation account remains an explicitly named precursor.

The modern word *interoception* should not be backdated to those sources. Their functional contribution was to connect body prediction, feeling, distributed reception, and action before the later vocabulary was adopted.

7.4 2017: the measurement program and claim ceiling become explicit

An owner-dated July 2017 note supplies an intrinsic receiver-state precursor. It describes a cell registering what it receives through changes in its own ionic condition, within a sequence of tonic cycling, burst, inhibition, emitted signal, returned response, and expectation [62]. The useful claim is physical registration by a changed receiver. The paper does not infer from this that one isolated neuron has a human emotional experience, and it does not import the note's stronger remote-field conjectures.

The Neural Lace work proposed synchronized study of brain activity, heart rate, pupil, gaze, movement, behavior, and environment [21]. A separate April 13, 2017 public episode gives the measurement proposal in operational form: align EEG with heart rate, eye tracking, pupil dilation, synchronized events, and machine pattern recognition rather than treating EEG as sufficient [68]. The same public program is independently triangulated by a 2017 Neural Lace compilation that joins the person, environment, brain interface, cardiac and eye variables, and self-report [93]. Its local header, displayed platform date, and first recovered 2018 Wayback capture remain separate. The exact speaker attribution of the Episode 3 passage still requires final audio reconciliation, so that passage is a provisional architecture source rather than a Micah-verbatim quotation. A June 2017 account also describes the brain as a special recording and playback medium in which thoughts and emotions can be related to changing signals [35].

More importantly, a November 2017 public discussion corrected an overcompressed phrase. Thoughts and feelings are **reflected in brainwaves**; they are not identical to brainwaves [22].

That distinction governs the present paper. Electrophysiological changes can be components and measurements of an emotional transition. The emotion remains a distributed brain-body event whose conscious form depends on how the transition is received, integrated, rendered, and used.

7.5 2018-2020: public geometry, decoding limits, and the person-level record

An August 2018 public Neural Lace episode proposed combining EEG, cardiac and muscle measures, eye and pupil behavior, movement, environment, and machine learning to study emotion and intention in interactive settings [86]. The episode header dates the recording to August 2; YouTube fixes the public upload on August 4. Its useful contribution is not the promise of effortless emotion reading. It is the public commitment to synchronized, multimodal, context-aware measurement before the later person-relative assay was formalized.

A September 24, 2018 public SVGN article then states that feelings and emotions are patterns in time and space: they have beginnings and endings, rises and falls, and shapes through which one experience can be related to another [69]. This is not yet a neural decoder. It is a public conceptual bridge from a named emotion to a measurable trajectory with onset, duration, magnitude, and bodily or spatial distribution.

The provenance is unusually strong but must still be stated precisely. The article displays a 2018 publication date and was independently captured by the Internet Archive on July 19, 2019. The same wording was later fixed inside the SAN Git repository in 2022. The emotion passage belongs to the September companion article, not to the May 2018 article whose material surrounds it in the local source file.

A May 2019 public episode provides the complementary claim ceiling. In Micah-attributed turns, a simple surface-signal account of reading emotion or thought is rejected because receiver state, dendritic integration, coincidence, and distributed transformation are being omitted [87]. Guest statements about biofeedback and mood remain guest material. The Micah-authored delta is that measurement is a partial projection of a more complicated receiving and transforming system.

A May 2020 public episode extends the design into a synchronized, longitudinal person-level record. Micah-attributed turns propose aligning heart, EEG, eye tracking, optical measures, head and body state, task events, and context; compressing recurring relations with machine learning; and protecting the resulting “digital double” as sensitive personal information [88]. This is a direct ancestor of the current person-specific calibration and privacy requirements. The three podcast transcripts were fixed in Git in October 2022, but the public audiovisual records supply their earlier 2018, 2019, and 2020 platform dates.

7.6 2021: emotion becomes baseline-relative and multiscale

An April 2021 Google Recorder discussion contains the clearest pre-PWD emotion bridge. In separately attributed turns, I asked whether emotion was necessary for decision and whether it operated from neurons to microcircuits to the whole brain. I asked whether fear, anxiety, and joy might correspond to detectable clusters of frequency-pattern changes marking movement toward a different inner state. I then proposed that an emotional state could be understood relative to alternative states or a baseline [23].

Other speakers in the same discussion contributed important distributed and physiological accounts. Those contributions remain theirs. The Micah-authored delta was the developing link among multiscale dynamics, baseline-relative difference, decision, and a mechanically specifiable emotional state.

The attribution boundary is concrete:

Recording interval	Speaker treatment in this paper	Use
00:03:52-00:04:49	Micah turn	Emotion, decision, and neuron-to-microcircuit-to-whole-brain scale
00:08:06-00:09:00	Micah turn	Fear, anxiety, and joy as candidate electrophysiological state transitions
00:17:10-00:18:12	Micah turn	Emotional state defined relative to alternatives or a baseline
Other intervals	Separately attributed participants	Context only; no migration into Micah-authored atoms

A public March 2021 source adds the instrumentation and ethics bridge: EEG, gaze, body tracking, heartbeat, breathing, affect inference, and privacy risk are treated as parts of a multimodal program rather than a universal mind-reader [36]. A same-day Internet Archive capture now supplies public custody for that article, strengthening the source from an owner-dated / Git-only route to a public archive-bound source.

Five recovered source recordings move parts of the mature mechanism into 2021 at the recording stage. An August 2 recording already describes an event shifting members of an oscillator group by milliseconds relative to their normal relation. Because the group is physically altered and re-equilibrates as a distributed system, the group itself performs the registration operation rather than sending the event to a separate inner observer [99]. A September 7 recording joins EEG-contingent light and sound with learned discrimination of changes associated with thoughts and feelings, then distinguishes spatial-temporal phasic patterns from tonic awareness [94]. An October 3 recording describes feeling as a causal structure that emits, contrasts, and detects its own distributed patterns [95]. An October 4 recording proposes feelings as spatial, temporal, and phase patterns in a body- and world-relative coordinate structure and compares that representational problem across feeling, sound, smell, and other senses [100]. An October 16 recording then expands the affective description through onset, peak, decay, scale, body location, spatial form, temporal form, and phase characteristics inside a responsive body-self-world map [96].

The embedded media timestamps bind the recordings, not their publication. Their public transcript fixations remain in 2022, and their existing transcript wording has not yet been manually aligned word-for-word to the audio. They therefore establish a strong 2021 source stage without being misrepresented as public 2021 quote custody.

An October 2021 recording returned to Neo Mind Cycle: EEG-derived light and sound re-entered through the senses, changed the next brain state, and expanded awareness of thoughts and feelings [24].

Another October recording contains repetition, deviation, cross-layer transformation, feeling, coincidence, and synchronization [37]. Its 2022 cleaned transcript adds explicit Shannon, tonic, and phasic language that has not been confirmed in the raw 2021 audio. This paper therefore uses the 2021 audio only for what can be heard there and treats the later cleanup as a separate 2022 formulation.

7.7 2022: the operation receives its mature SAN names

The 2022 record is not one leap from an old metaphor to a finished acronym. It contains a recoverable sequence.

At the measurement layer, a June 2022 public Git source proposes synchronizing heart rate, EEG, eye tracking, pupil dilation, behavior, and artificial intelligence to estimate a person's changing emotion and intention [57]. The note embeds wording that it attributes to 2012, but the hard public fixation used here is June 8, 2022. It is a partial ancestor of the present multimodal assay, not evidence that the later held-out estimator, perturbation controls, or causal rescue design already existed.

The second-pass audit recovered an earlier June public spine that had been underemphasized. One note says that the brain maps changes in its own electrical oscillations and chemistry while synchronizing maps of feeling, knowing, sensation, action, and breathing [70]. Another makes the tonic-canvas / phasic-ink distinction information-theoretic: a recurring background carries less new information than a difference that changes the current pattern [71]. A third joins feelings, phase dispersion, detected disturbance, memory, context, and a distributed self-model [72]. A fourth explicitly proposes that emotion and qualia are detected as distributed phase-pattern differences, while leaving its specific fear-loop anatomy as a hypothesis rather than an established map [73]. Together these sources publicly fix the self-map, background / difference, distributed-detection, and affective-rendering chain by June 2022.

A June 11 source adds a slower learning timescale. It proposes that neuronal firing changes mitochondrial ATP-related activity, protein and receptor growth, future recurrent-network learning, and the refinement of choice [101]. This is used here as a historical SAN metabolic-plasticity hypothesis: it supplies a route by which earlier state and action history could change a receiver's future condition. It does not establish that mitochondria encode emotion, that the proposed electromagnetic coupling is the accepted causal route, or that the complete later PWD-NAPOT-emotion architecture was already fixed on that date.

Two public videos independently preserve the same architecture in spoken form. On July 31, the one-hour NAPOT explanation states that each neural array receives only part of a rendered picture, that oscillating cell groups perform the observer operation, and that feelings and emotions belong to the proposed volumetric rendering [89]. On September 18, the public NAPOT Revision 5 video describes a maintained baseline of experience, departures that alter rendered content, and the brain detecting its own information through distributed feedback; it also presents emotional change against tonic coordination as an explicit conjecture [90]. These are distinct public-platform sources, not earlier byte fixations of later Git notes. Their public upload dates, repository-link dates, local transcript derivatives, and later Git wording remain separate.

First, a0193z states that when one neuron departs from its regular oscillatory relation, that departure produces a temporal difference detectable by the receiving group [38]. a0295z then distinguishes phasic events from tonic expectation and proposes that smaller differences can contribute to ordinary perception while larger departures can recruit stronger attention [39]. This is a graded-difference hypothesis, not a law that salience always means one high frequency.

Second, a0171z joins structured excitation and inhibition, phase change, sensation, feeling, prediction, language, and muscular action [40]. a0306z describes a content-bearing phase-pattern departure detected by dendritic receivers and transformed into downstream rendering; it applies the same receive-transform-project operation to color and to the cross-modal, temporospatial properties of broccoli [41]. a0001z places thought sequences, emotion, and feeling inside the rendered body model and later applies the receiver-relative account to the experience of redness [42].

Several 2022 sources also make the geometry of feeling explicit. A June transcript describes onset, peak, magnitude, duration, rise and fall, and felt body location as separable properties of an emotional episode [75]. A July note proposes learned phase-sequence geometry as a representation of those properties [80]. A September note distinguishes the rendered feeling of being somewhere in one's body from a tonic condition with no intrinsic location [74]. These claims do not imply that a feeling is one literal geometric object. They motivate measuring its time course, body map, route, and changing spatial distribution instead of reducing it to one category or scalar.

The no-inner-observer chain is equally direct. A June Git-fixed note says that the brain's organized regions detect large-scale changes in oscillatory flow and render those changes as feelings [77]. A separately preserved transcript says that feeling is the machine detecting its own causal structure: the same distributed structure emits, contrasts, and detects the relevant patterns [76]. The latter is now bound to an October

3, 2021 recording-service timestamp and a June 2022 public Git transcript fixation. Those dates remain distinct, and the derivative transcript is still marked as not manually aligned word-for-word to the audio. A first-person neurofeedback note adds phenomenological support for learned discrimination of brainwave, cardiac, and bodily change, but is not treated as proof of a unique neural generator [78].

Third, the August 2022 Whitepaper joins feeling and emotion memory, tonic context, phasic departure, choice, embodied arrays, and volumetric phase rendering in one public architecture [43]. The Neo Mind Cycle bridge is retained explicitly as a mirror of the brain to itself: measured activity changes the display, returned sensory information changes the next brain state, and the next state changes the next measurement [55].

An October Whisper transcript then states the bridge in one continuous first-person account: EEG-derived feedback can teach a person to discriminate changing thoughts and feelings, while emotions are modeled as spatial-temporal phasic differences against ongoing tonic awareness [79]. Its original Google Recorder audio is available, but the wording cited here is the public 2022 Whisper transcript; retrospective statements about 2011 practice are not used to backdate the mature tonic / phasic terminology.

By November, SAN described an ongoing context maintained by recurrent neural activity, event-related departures from that context, distributed receivers and transmitters, and a sensorimotor rendering that included thoughts, feelings, and emotions [25,26]. The emotion conjecture proposed that a large emotional change might appear as a reconfiguration of cooperating brain regions and their coordinated dynamics. Its older shorthand associated emotions too directly with named frequency bands. The stronger formulation already present in the same genealogy is multivariate and relative: an emotion is a transition in a distributed state measured against that person's preceding context.

Another November source makes the observer claim emotion-specific. It says that emotional processing is whole-brain but unequal: each receiving group responds according to its learned dendritic and synaptic configuration. It then describes experience as a cascading loop of detected renderings and asks what observes a generated feeling. Its answer is the same distributed machine that generates and is changed by the feeling [58]. This is the clearest public 2022 bridge from receiver-specific emotional processing to self-registration without an inner reader.

The joined November 2022 NAPOT 6 source then places the pieces in one chain: differences against tonic activity become information; feedback-bound differences contribute to emotion, thought, and interior self; oscillating cell groups operate as observer-actor systems; learned patterns help select action; and returned sensation modifies what happens next [63]. Its broad band and dominance language remains a SAN hypothesis. The durable contribution here is the joined receive-differentiate-render-act-return architecture.

The same year introduced an explicit valence bridge. Psychological valence described appetitive or aversive direction. A proposed electrical form described how a recruited pattern could gain greater effective influence through receiver-specific currents, timing, and propagation [27]. The shared word *valence* does not make these variables identical. Psychological value can change which assemblies are recruited; the electrical mechanism concerns how a recruited assembly affects downstream receivers.

7.8 2023-2026: public synthesis and testable specialization

The later record strengthens continuity, but its authorship and custody layers must remain visible:

Date and source	Contribution to the emotion mechanism	Attribution and custody boundary
2023-12-13, SAN Whitepaper Draft 3 [44]	Places feeling and emotion in phase-pattern and volumetric rendering; identifies the oscillating collective as observer and tonic activity as canvas.	Public Git custody; explicitly includes ChatGPT 4 text. Constituent earlier notes govern Micah-verbatim priority.
2024-01-09, a0254z [59]	Connects the tonic-versus-difference architecture to information theory: homogeneous synchrony carries little new information, while relation-changing deviations can alter timing, encode a new pattern, and contribute to a new affect.	Micah-verbatim public Git source. It rejects synchrony alone as sufficient, but does not establish a universal band-to-emotion map.
2024-01-22, direct Micah block in GPT2 024Jan1toJan22.md [81]	Joins low-information tonic readiness, arising rendered forms, semantic segmentation, and attached valence.	Only the bounded Micah-authored block is used; surrounding assistant text is excluded from authorship evidence.
2024-01-23, a0403z [45], and 2024-01-24, <i>Cortical Dynamics</i> [28]	Joins emotional context, hedonic valence, arousal, salience, visceral input, autonomic state, value-based choice, and recurrent cortical-subcortical loops.	Micah-published and curated; anatomy claims require independent verification. The article has a 2024-03-02 Wayback capture.
2024-09-06, 02san.md [46]	Direct Micah turns connect thoughts, feelings, desires, tonic context, PWD, derivative-like mismatch, and changing inner experience.	Dialogue source; Micah turns and assistant replies remain separate.
2024-09-07 to 2024-10-22, three SVGN syntheses [47,48,49]	Extends feedback, tonic / PWD, brain-state, and distributed-observer architecture.	AI-assisted or explicitly AI-composed around Micah prompts and source notes.
2024-10-06 predictive-coding comparison [98]	Publicly stated that PWDs encode prediction error and that a compared study supplies empirical support for SAN.	Historical public claim preserved as challenged. This paper rejects both the definitional identity and the resemblance-to-validation shortcut.
2024-10-24 book and 2024-10-30 announcement [50,51]	Publicly joins thoughts, feelings, emotions, tonic / PWD variation, and a dynamic three-dimensional neural workspace.	Micah publication with substantial AI assistance. The Git PDF and review PDF are different editions; the original announcement capture governs what was public in 2024.
2025-01-01, <i>A New Law of Thermodynamics</i> [97]	A bounded first-person Micah block joins high-phasic PWD dissipation, synchronized tonic receivers, stable rendering, and movement toward a changed equilibrium.	Late public operator bridge, not an earlier origin. The current page and platform timestamp are available, but an immutable January body capture is not; surrounding explanatory prose is not treated as Micah-verbatim.
2025-04-08 Friston presentation and 2025-06-02 article [52]	Connects iterative difference reduction and mismatch dissipation to the broader SAN architecture.	Micah talk and publication; general mechanism bridge rather than an emotion-specific origin.
2025-02-17 and 2025-06-15 bounded Git sources [82,83]	Distinguish body-origin from intra-brain streams and propose a coefficient-of-variation discriminator for phase-pattern difference against collective background.	Direct Micah blocks only. The coefficient is a candidate measurable statistic, not an established emotion biomarker.
2025-08-21 SAN / OCA repository record [53]	Supplies public tonic / phasic, embodiment, active-inference, and distributed-observer context.	Zenodo's hard record date is 2025-08-21 even though metadata declares 2025-05-16; emotion coverage is thin.
2026 governed neuroaffective specification [54]	Joins the full brain-body transition, person-relative estimator, feedback controls, privacy constraints, and decisive causal tests.	Current mixed-assistant governed synthesis; not a historical-priority source or an empirical result.

7.9 What the 2011-2026 coverage does and does not establish

The genealogy is continuous as a developing research program, but it does not contain one equally strong source for every calendar year. That distinction is part of the result:

Year	Recovered role	Date and authorship boundary
2011	Physical sense-state-action-return loop, collective-cell self, and weighted choice ancestry	Owner-dated wording; public Git fixation begins in 2022
2012	External self-mirror, prediction-based emotion, distributed observation, physical-form / learned-meaning distinction, regulation, and action	Owner- or source-dated wording; public Git fixation begins in 2022
2013	Predicted fear, no-inner-observer architecture, dynamic coincidence-pattern self, and action criteria	Owner-dated wording; public Git fixation begins in 2022
2014	A possible multisensory self-pattern and body-feeling bridge	Claimed date remains source-gated and is not used as a priority anchor
2015	Body-sensation prediction and self-referential conscious feeling	Owner-dated wording; public Git fixation begins in 2022
2016	No nonredundant emotion-mechanism source recovered in the bounded pass	Explicit no-hit; no source or date is manufactured
2017	Public multimodal measurement program and “reflected in, not identical to, brainwaves” claim ceiling	Public episodes plus a separately owner-dated cellular receiver precursor
2018	Public multimodal emotion / intention program and spatial-temporal geometry of feeling	Public video and article dates
2019	Public rejection of simple surface-signal thought or emotion reading	Public video date; automated transcript fixed in Git in 2022
2020	Public synchronized longitudinal person-level measurement and privacy architecture	Public video date; automated transcript fixed in Git in 2022
2021	Multiscale, baseline-relative emotion, closed-loop self-discrimination, causal self-detection, and affective geometry	Public / archive-bound articles plus service-metadata recordings; later transcript fixation and speaker boundaries remain separate
2022	Mature public receive-differentiate-render-act-return architecture, tonic context, PWD, affective geometry, and self-observation	Public Git, recordings, and videos; exact bytes remain bound to their own fixation dates
2023	Public Whitepaper synthesis	Public Git; assistant-composed text is not silently attributed to Micah
2024	Public information-theoretic, neurocircuit, wave-life, phase-field, and book synthesis	Public Git, SVGN / Wayback, and book custody with assistance disclosed where applicable
2025	Friston bridge, body-origin / intra-brain distinction, candidate dispersion statistic, and OCA record	Public talk, Git, article, and Zenodo dates remain separate
2026	Person-relative causal assay and the present formal specialization	Governed unpublished synthesis; not historical or empirical proof

The current paper specializes that broad architecture. Its object is neither a general consciousness theory nor a generic emotion classifier. It is the mechanism and experiment by which a distributed brain-body system registers its own affective transition. The chronology supports continuity; it does not turn later terms into earlier wording or AI-generated passages into unaided Micah-verbatim text.

8. What the receiver-relative model adds beyond inherited descriptions

The positive mechanism comes first. Its relationship to inherited concepts can now be judged precisely.

8.1 PWD is not a synonym for phase coding, traveling waves, or prediction error

Inherited concept	What it already explains	The additional operation proposed here
Phase coding	Information can depend on event timing relative to an oscillatory cycle.	The difference is receiver-relative and multivariate, including expected relation, duration, waveform, magnitude, route, and downstream consequence.
Traveling waves	Activity propagates across tissue with measurable phase gradients and direction.	Propagation becomes one carrier inside a receive-transform-project-return loop; a traveling wave alone is not self-registration.
Interoception	The nervous system senses and integrates internal bodily signals.	The paper specifies distributed registration as receiver state change and joins it to tonic context, PWD, rendering, action, and external-loop tests.
Predictive or active inference	Predictions and errors organize perception, bodily regulation, and action.	The paper operationalizes the candidate error at multiple receivers and requires phase-, duration-, and route-sensitive incremental prediction and causal rescue.
Constructed emotion	Context, concepts, bodily signals, and prior learning contribute to emotion.	The paper specifies an electrotemporal and recurrent mechanism by which the changing system registers its own state without an inner interpreter.
Affective prosody and emotional semantics	Physical vocal cues and learned linguistic meaning can make separable and interacting contributions to neural, autonomic, and reported responses.	Equation [1] makes the event receiver- and history-dependent, and Experiment 6 tests whether physical form, learned meaning, and their interaction alter later self-registration through distinguishable routes.
Neurofeedback	A measured signal can be transformed and returned for learning or regulation.	The external loop is used as a controllable surrogate for testing the hypothesized intrinsic self-registration loop.
Global workspace	Selected information can become widely available for report and control.	Availability is implemented by many receiver-transmitters whose state changes constitute distributed observation rather than a central reader.

One earlier SAN comparison article overcompressed this relationship by saying that PWDs encode prediction error and that a predictive-coding result provides empirical support for SAN [98]. Neither shortcut is adopted here. A prediction error may be one kind of receiver-relative difference in a declared model, but Equation [5] also permits differences in timing, waveform, magnitude, duration, route, and relation that are not prediction errors by definition. Likewise, an external result becomes evidence for this proposal only when it tests the declared SAN variables and discriminators; conceptual resemblance alone is not empirical validation.

The distinction is empirical. If SAN variables add no prediction or selective causal leverage after strong versions of these models are implemented, the emotion-specific SAN addition fails.

8.2 Division of explanatory labor across companion papers

This paper shares several SAN mechanisms with companion manuscripts, but it does not present their specialized contributions as new results of this paper.

Companion paper	Its primary explanatory responsibility	What This paper imports	What remains distinctive here
<i>The Coherence-Dispersion Sculpting Hypothesis, The Coherence-Dispersion Sculpting Hypothesis</i>	Qualitative differentiation through local assembly coordination, structured phase separation, altered-state dynamic range, and plasticity.	The possibility that coordinated and separated population relations can contribute to a receiver’s changing context.	Emotion-specific brain-body transition, intrinsic self-registration, affective trajectory and body-map measurement, and physiology-feeling dissociation tests. this paper does not claim that an emotion assay establishes a general mechanism of qualia.
<i>Self-Aware Networks: A Systems Whitepaper, Self-Aware Networks: A Systems Whitepaper</i>	The complete SAN receive-transform-project-complete-observe-act-return architecture across perception, memory, embodiment, and artificial systems.	The distributed-observer, tonic / phasic, PWD, embodied-action, and returned-consequence framework.	One domain-specific specialization with a 2011-2026 emotion genealogy, person-relative measurement model, and six discriminating experiments. this paper does not restate the entire SAN architecture or claim that its emotion tests prove NAPOT.
<i>Coherence-Weighted Electrical Drive in Choice and Motor Realization, Coherence-Weighted Electrical Drive in Choice and Motor Realization</i>	A signed, receiver-specific effective-drive estimator, candidate-route commitment, motor realization, and returned-consequence tests.	The requirement to keep psychological valence, electrical drive, phase, power, and action distinct.	Registration of a changing emotional condition before, during, or without motor commitment. this paper does not define <i>Coherence-Weighted Electrical Drive in Choice and Motor Realization</i> ’s drive estimator or treat action as necessary for every consciously accessible feeling.

The fan-out is therefore directional rather than duplicative. this paper can supply person-relative affective states, transition estimates, and dissociation cases as candidate inputs or test domains for *The Coherence-Dispersion Sculpting Hypothesis*, *Self-Aware Networks: A Systems Whitepaper*, and *Coherence-Weighted Electrical Drive in Choice and Motor Realization*. Those companion models can in turn supply stronger content, architecture, or motor hypotheses for this paper’s nested comparisons. A positive result in one paper does not automatically validate the distinct claim owned by another.

9. Experimental program

Experiment 1: person-relative transition mapping

Recruit a preregistered, adequately powered sample and elicit bounded affective transitions using validated stimuli, autobiographical recall, social feedback, and neutral controls. Synchronize:

- high-density EEG;
- ECG or PPG and heart-rate variability;
- respiration;

- electrodermal activity;
- facial and postural EMG;
- pupil and gaze;
- skin temperature;
- head and body movement;
- task events;
- time-resolved body-location maps and episode-onset, peak, duration, and rise / fall judgments; and
- continuous, event-locked, delayed, and retrospective self-report conditions.

Compare:

1. unimodal models;
2. fused conventional models;
3. fused models plus preregistered receiver-relative SAN features;
4. participant-specific models;
5. cross-person models; and
6. held-out-session, held-out-time-block, and held-out-person performance.

Use nested validation grouped by participant, session, emotional episode, and time block. Autocorrelated windows from one transition must not cross folds. Feature selection, frequency bands, time windows, scaling, harmonization, and hyperparameters are fitted inside training data only. Report discrimination, calibration, test-retest reliability, transition-onset timing, and uncertainty, not accuracy alone. Conventional affective-computing datasets such as DEAP may serve as preliminary benchmarks but cannot substitute for a causally designed person-relative protocol [29].

EEG phase analyses must control source leakage, volume conduction, ocular and facial-muscle activity, respiration, movement, filter-induced phase distortion, and circular selection of bands and windows. Repeated trials from one participant must not leak across train and test folds. Targets must be declared separately: arousal, pleasantness, a named category, action readiness, reported intensity, and regulation success are different outcomes. Because continuous report itself changes attention and state, compare it with event-locked, delayed, and retrospective report rather than treating one method as a neutral readout.

Test affective geometry as an incremental target, not a decorative visualization. Ask whether the time-resolved neural and bodily features predict held-out onset, duration, spread, contraction, and body-location judgments after ordinary arousal, valence, category, movement, respiration, and person identity are controlled. A geometry claim fails if it is unreliably reported, adds no prediction beyond those baselines, or changes only because the reporting interface directs attention to a location.

Experiment 2: external self-mirror causality

Transform selected features into light, sound, or bounded XR changes. The full control set is too large for one indiscriminate eight-arm study; use a staged factorial program while preserving these contrasts:

- veridical contingent feedback;
- fixed-delay feedback;
- replay from the same participant;
- replay from another participant;
- decorrelated feedback with matched spectrum and magnitude;
- phase-scrambled feedback;
- feedback with matched positive or negative valence but no contingency; and
- no-feedback control.

Match spectrum, displayed magnitude, display latency, sensory complexity, reward schedule, instructions, and expectancy wherever the contrast permits. Advance from a smaller feasibility stage to a powered confirmatory stage with the mapping and primary endpoint frozen.

The primary question is not whether participants enjoy the display. It is whether true contingency supports learned discrimination or regulation beyond the effects of expectation, sensory stimulation, task engagement, and feedback valence.

Experiment 3: physiology-feeling dissociation

Use ethically approved perturbations and naturally occurring variation to identify:

1. physiology changes without the predicted reported feeling;
2. reported feeling changes without the predicted signature in the measured modalities;
3. equivalent arousal with different valence;
4. equivalent valence with different arousal;
5. equivalent movement with different reported affect; and
6. equivalent report with different person-specific trajectories.

The model succeeds by explaining dissociation, not by excluding inconvenient trials.

Experiment 4: receiver-relative causal perturbation

Identify a preregistered circuit and phase / timing relation that predicts the target after ordinary power, rate, movement, respiration, and history are matched. Use three distinct evidence tiers rather than pretending that one generic human protocol can manipulate every receiver:

1. non-invasive feedback or stimulation with explicit direct-sensory controls;
2. opportunistic intracranial human observation or stimulation where clinically justified and independently governed; and
3. invasive circuit intervention in an appropriate animal model when the mechanistic question cannot be answered safely in humans.

Dynamic causal modeling or another predeclared causal model may organize candidate directed relations, but model fit is not intervention proof [30]. Require the predicted selective loss, then test a matched rescue that restores the relation without restoring the nuisance variables.

Experiment 5: intrinsic versus external registration

Compare three conditions:

1. no external display, while ordinary intrinsic bodily and neural return remains present;
2. external feedback that accurately mirrors a selected internal variable; and
3. external feedback that conflicts with the internal variable.

Measure which source dominates report, action, learning, and later expectation. This directly tests whether the external Neo Mind Cycle mirror reveals, amplifies, competes with, or reshapes an already operating intrinsic loop.

Experiment 6: physical form, learned meaning, and emotional transition

Use a preregistered within-person staged design that separates what a stimulus is physically from what it means to the participant without treating undefined or derived variables as independent factors:

1. **Understood-language lexical block.** Cross neutral versus affective prosody with emotionally neutral versus valenced semantics. Treat prosody-meaning congruence as a derived contrast from this crossing, not as a fifth independent factor.
2. **Sensory and lexicality block.** Compare acoustically controlled familiar words and pseudowords while manipulating prosody. Estimate lexical-semantic effects only for stimuli for which meaning is defined.
3. **Language-history block.** Compare understood and unfamiliar languages under matched acoustic controls while explicitly measuring comprehension, borrowed-word recognition, speaker identity, and cultural familiarity.
4. **Learned-cue block.** Train arbitrary sound-meaning associations, then reverse or retrain them. This tests receiver history without pretending that established natural-language meaning can be reassigned in one session.

Validate acoustic duration, intensity, fundamental frequency, spectral content, and intelligibility independently. Do not treat an unfamiliar language as meaningless if the participant recognizes borrowed words, speaker identity, or cultural cues. Measure trial-level EEG or MEG where available, ECG or PPG, respiration, electrodermal activity, facial EMG, pupil, movement, continuous or delayed feeling report, body-location report, and later memory or action bias.

Fit a preregistered hierarchical model with block-specific estimands. Estimate separately:

- a physical-form effect that generalizes across learned meaning;
- a learned-semantic effect after matched sensory structure;
- their interaction;
- modification by the participant’s language and learning history; and
- whether early changes predict later interoceptive report, regulation, or choice after ordinary acoustic, autonomic, and contextual baselines.

Train all scaling and feature selection inside participant-grouped folds. The result supports the proposed extension only if receiver-history variables add held-out prediction or selective causal leverage beyond physical stimulus features and conventional semantic, appraisal, and autonomic models. No word, sound, prosodic pattern, or language is assumed to have a universal emotional meaning.

Ethics, privacy, and intended use

The same multimodal data that support self-regulation could be misused for covert emotion inference. The 2021 source program already identified that risk [36]. The protocol therefore requires informed consent, distress screening, withdrawal without penalty, medication and sleep covariates, adverse-event monitoring, encrypted participant-controlled data, and a prohibition on covert workplace, insurance, policing, or interpersonal emotion scoring. A person-specific model is not a warrant to infer an unconsenting person’s mental state.

Claim-to-test map

Atom	Observable	Strong baseline	Intervention or contrast	Failure condition
NEA-01	Declared event, physical stimulus structure, learned meaning, and transition onset	Matched neutral / task event plus staged form / meaning controls	Validated affective event and Experiment 6 block-specific contrasts	No transition beyond task, physical salience, learned meaning, or expectancy

Atom	Observable	Strong baseline	Intervention or contrast	Failure condition
NEA-02	Multimodal neural and body trajectory, including episode timing and reported body map	Unimodal and nuisance-complete fusion	State induction or regulation	No reliable multimodal structure
NEA-03	Sensor-specific projection and reliability	Artifact-only and identity-leakage models	Sensor ablation	Result requires treating one projection as the emotion
NEA-04	Recurrent and interoceptive return	Exteroceptive-only and delay models	Delay or conflict feedback	Return adds no prediction or leverage
NEA-05	Standardized, circular, waveform, route, and relational discrepancies conditioned on receiver history	Absolute power, rate, raw stimulus features, and history-aware recurrent model	Timing, route, or learned-association perturbation	No nested held-out increment
NEA-06	Receiver change with downstream consequence	Central-decoder and non-recurrent models	Receiver or return-path perturbation	Distributed registration has no explanatory gain
NEA-07	Separate feeling, awareness, report, and affective-geometry endpoints	Physiology-only, report-only, context-only	Dissociation and display conflict	Model requires feeling-physiology identity or a universal geometric fingerprint
NEA-08	Value, accessibility, effective drive, and action	Value-only and drive-only policies	Value / timing manipulation	Cannot separate evaluation from recruitment
NEA-09	Returned consequence and learning curve	Sham, delay, replay, decorrelation	True contingency	Contingency adds nothing beyond matched controls
NEA-10	Person-specific calibration, prediction, perturbation, association reversal, and rescue	Cross-person and conventional multimodal models	Staged full protocol including Experiment 6	Neither reliable increment nor selective causal leverage

10. Predictions

1. Person-specific models will initially outperform universal emotion classifiers.
2. Multimodal models will distinguish some transitions that EEG-only and peripheral-only models miss.
3. The gain from fusion will depend on task, person, time scale, and target.
4. Receiver-relative features will add value only in selected circuits and windows, not uniformly across the brain.
5. True contingent feedback will produce learning effects distinguishable from matched sham, delay, replay, and valence effects.
6. Some autonomic changes will occur without corresponding reported feeling, and some similar reports will arise through different trajectories.
7. A successful SAN feature will predict not only a label but a later change in report, regulation, action, or learning.
8. Selective disruption of a genuinely causal receiver-relative relation will impair the predicted component, and a matched rescue will restore it.
9. Some person-specific affective-geometry features will predict transition and regulation targets beyond scalar arousal and valence, but no universal geometric fingerprint will succeed for every person and emotion.

10. Physical stimulus structure and learned meaning will make separable contributions in at least some tasks, and their interaction will depend on the participant's language, learning, and current context.
11. A valid receiver-history effect will generalize to independently collected sessions or change predictably after retraining or reversal; it will not be explainable solely by uncontrolled acoustic salience or participant identity.

11. Failure conditions

The emotion-specific SAN proposal should be revised or retired if:

- conventional multimodal models explain held-out data as well as or better than the SAN extension;
- receiver-relative features fail across independently collected sessions;
- their lower-bound improvement does not exceed the preregistered minimum or fails independent replication;
- gains disappear after movement, respiration, facial-muscle, task, and identity leakage are controlled;
- feedback effects are reproduced by matched sham, valence, replay, or engagement conditions;
- selective perturbation does not produce the predicted component-specific change;
- rescue does not restore the targeted function;
- alignment and dispersion variables become uninterpretable after circular, source-leakage, and route controls;
- onset, peak, offset, duration, intensity, and body-location features are not reliable within person across independently collected sessions;
- affective geometry adds no held-out prediction or causal leverage beyond conventional body maps, episode timing, arousal, valence, movement, respiration, task, and context;
- the model requires one universal frequency-emotion mapping that the data do not support;
- the physical-form, learned-meaning, and interaction terms cannot be separated from acoustic salience, task demand, familiarity, or expectancy;
- receiver-history variables add no reliable prediction and do not change after preregistered association training or reversal; or
- the proposed variables cannot be measured with acceptable reliability.

These are productive failure conditions. They separate a mechanism from a metaphor.

12. Discussion

The source genealogy reveals a coherent research path. The 2011 work supplied a physical self-loop, placed feeling and action inside its returning pathway, and described the experienced self as a collective of cells. The 2012 work built an external mirror of changing brain activity and related it to thought, feeling, recognition, and regulation; it also made emotion prediction-based and joined dynamic electrical expectation to brain-body-sensory feedback and distributed observation. It additionally separated a stimulus's physical form from its learned meaning and made receiver history part of the causal question. The 2013-2015 work added criteria, distributed cellular reception, dynamic self-reconstruction from coincidence, and the absence of an inner witness. The 2017 work added a changed-receiver registration precursor, made the public multimodal program explicit, and corrected identity language. The 2018 public record added both multimodal interactive measurement and the tempo-spatial geometry of feeling; the 2019 and 2020 public episodes added the limits of simple decoding and the synchronized person-level record. The 2021 work framed emotion as a multiscale, electrophysiologically approachable transition defined relative to other states or a baseline and, in service-dated recordings, joined tonic / phasic self-discrimination, causal self-detection, and affective

geometry before their later public transcript fixations. The 2022 public record added a synchronized multimodal emotion-and-intention assay ancestor, tonic context, PWD, neural rendering, affective trajectory and body location, valence, action, an explicit account in which the feeling-generating machine is also its observer, and a joined observer-act-return NAPOT operator. Public 2022 explanatory videos independently preserve the partial-projection, distributed-observer, tonic-context, and rendered-emotion architecture. The 2023-2025 public sources integrated those operations into the Whitepaper, an information-theoretic tonic-versus-difference account, cortical-loop, book, mismatch-reduction, and repository record. The governed 2026 synthesis supplied the person-relative causal assay. The present paper converts that genealogy into a measurement and falsification program.

The paper's distinctive claim is not merely that emotion has physiology. It is that a distributed brain-body system can register an emotional transition by being changed at many partial receivers, compare those changes with maintained context, reintegrate them through recurrent and interoceptive return, and use the resulting state in feeling, regulation, choice, and learning without a central reader. That joined operation is stronger and more testable than a list of correlates.

The external neurofeedback mirror is scientifically useful because it makes one part of the loop manipulable. A machine measures a partial projection, converts it into a sensory consequence, and returns it. The intrinsic nervous system does something more complex but structurally related: internal receptors and neural receivers are changed by bodily and neural transitions; their outputs return through recurrent pathways; later processing changes. The analogy is not that the brain contains a literal screen. It is that both loops contain measurement, transformation, return, receiver change, action, and learning.

This architecture also clarifies what it means to say that a person can detect an electrophysiological shift paired with emotion. The person need not sense a voltage value as a voltmeter does. The distributed system is altered by neural and bodily currents, afferent signals, movement, chemistry, and context. Some of those changes become discriminable through interoception, attention, memory, and feedback. Conscious recognition is a developed capacity of the loop, not a separate reader looking at it.

13. Conclusion

An emotion is a distributed brain-body transition whose components unfold over different spatial and temporal scales. Measurements reveal partial projections. A feeling's onset, peak, decay, and reported bodily distribution are candidate projections of that transition, not a complete identity statement. Interoceptive and recurrent pathways return parts of the transition to a system whose present state determines what those signals can do. The resulting receiver changes alter feeling, prediction, action, and learning.

Self-Aware Networks proposes that this distributed causal alteration is the observer operation. Tonic context supplies the maintained condition; PWD names the receiver-relative differences; Neural Rendering names their distributed incorporation into the changing body-self-world model. The theory is scientifically useful only where these variables improve held-out explanation and survive selective perturbation and rescue.

The next step is therefore experimental, not terminological: measure the transition across the brain, body, action, and report; control the artifacts; manipulate the return; and determine whether the proposed receiver-relative mechanism explains what simpler models leave unresolved.

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Source-custody note

The detailed date, authorship, hash, and custody audit is maintained in:

- analysis/EMOTION-GENEALOGY-2011-2026-SOURCE-RECOVERY.md
- sources/emotion-genealogy-ledger.json
- sources/emotion-atom-coverage-matrix.json
- sources/emotion-genealogy-ledger-draft4.json
- sources/emotion-atom-coverage-matrix-draft4.json
- sources/draft4-source-review-corrections.json
- sources/draft4-genealogy-additions.json
- sources/san-git-emotion-census.json
- sources/san-git-emotion-candidate-dispositions.json
- analysis/pending-review-tranche-a-ranks-001-150.json
- analysis/pending-review-tranche-b-ranks-151-300.json
- analysis/pending-review-tranche-c-ranks-301-449.json
- validation/pending-review-tranches-aggregate-validation.json
- analysis/DRAFT4-EARLY-EMOTION-GENEALOGY-RECOVERY-2011-2021-20260725.md
- analysis/DRAFT4-LATE-EMOTION-GENEALOGY-RECOVERY-2022-2026-20260725.md
- analysis/DRAFT4-2022-PUBLIC-VIDEO-EMOTION-SOURCE-BINDINGS-20260725.md
- analysis/DRAFT5-RECORDER-RETRANSCRIPTION-AND-CONVERSATION-INTAKE-20260725.md
- sources/draft5-genealogy-additions.json
- sources/emotion-genealogy-ledger-draft5.json
- sources/emotion-atom-coverage-matrix-draft5.json
- tools/build_draft4_sources.py
- tools/build_draft5_sources.py
- tools/build_draft5_manuscript.py

Internal source dates, service metadata, Git fixation dates, publication dates, archive captures, and recovery dates remain separate. The deterministic Git census routes 449 candidate files. The first disposition layer accepted 33 paths, used two additional paths with explicit source gates, and left 414 paths pending. The second-pass tranche reports then screened all 414 pending paths exactly once with rank, path, and input-SHA alignment: 227 were deep-read, 187 received screening-level classifications, and 23 were flagged as material-addition candidates. Screening is not acceptance. Draft 3 promoted only attributable, source-bound, nonredundant findings. Draft 4 leaves that baseline immutable and records any corrected classification or outside-census addition in a separate overlay before adding a source row. Mixed, speculative, quoted-external, AI-synthesis, attribution-gated, and source-gated routes remain separately visible in the reports and dossier.

The Draft 4 ledger remains immutable at 111 atom-stage-source rows. Draft 5 adds three governed rows: the August 2, 2021 distributed-registration recording, the October 4, 2021 affective-geometry recording, and the June 11, 2022 metabolic-plasticity / choice Git source. The resulting Draft 5 ledger contains 114 rows. A row marked source-gated, private, mixed-assisted, challenged, or recording-stage is preserved for auditability; it is not silently promoted into a public-publication priority claim or empirical result.